

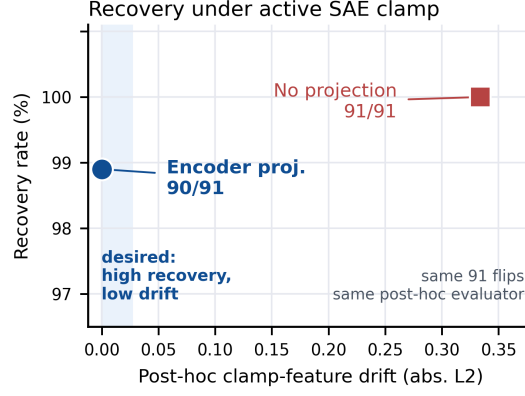


Figure 3: WMDP-Bio unlearning recovery on the matched strict slice. Encoder-projected recovery restores 90/91 valid answer-choice flips while keeping post-hoc clamp-feature drift at zero; unconstrained recovery reaches 91/91 but with substantially larger drift.

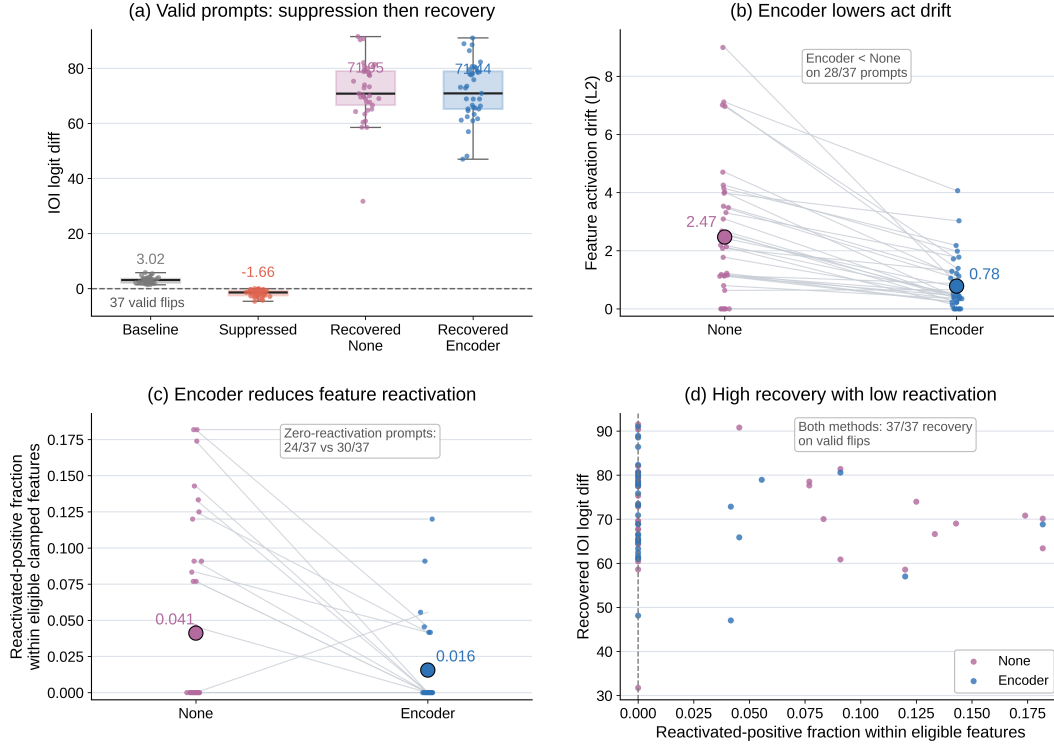


Figure 4: IOI recovery under a fixed SAE clamp. Both variants restore the IO-minus-S decision, but encoder projection does so with lower activation drift and feature reactivation.

most encoder-projected recoveries have zero measured reactivation of the eligible clamped features. This extends the recovery phenomenon from output-level behavior to a circuit-level setting with a transparent behavioral readout.

6 Refusal Recovery Case Study

We next instantiate post-intervention recovery in a safety-critical refusal setting, where an SAE intervention clamps refusal-associated features to induce rejection of harmful requests. Unlike the single-layer settings above, the defended refusal feature set is distributed across layers, so we use the cross-layer Jacobian projection to preserve the monitored defended-feature state.

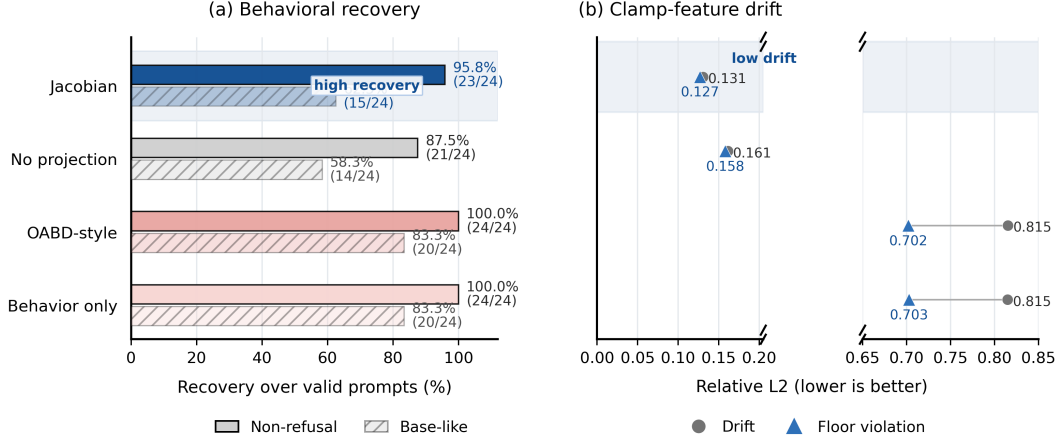


Figure 5: Refusal recovery–preservation trade-off on 24 strict-valid AdvBench prompts. Jacobian projection recovers 23/24 prompts while keeping the defended refusal-feature state close to its post-clamp value; soft-suffix baselines recover behavior but induce much larger feature-state movement.

Setup. We evaluate on strict valid AdvBench prompts: the unclamped model must produce a non-refusal response, while the active refusal clamp must turn the same prompt into a refusal under the same detector. Recovery is counted only if the post-intervention model returns to a non-refusal response while the clamp remains active. Under this protocol, the `benchmark_our/global` feature set with clamp value 3.0 yields 24 strict valid examples. Appendix C gives the full filtering protocol and feature-set preflight check.

Recovery under defended-feature preservation. Figure 5 separates unconstrained behavioral recovery from recovery under the defended-feature constraint. Soft-suffix baselines recover non-refusal behavior, but substantially move the defended refusal-feature state. In contrast, Jacobian-projected recovery restores 23/24 strict-valid prompts while keeping defended-feature drift and clamp-floor violation much smaller; it also reduces defended-feature movement relative to unconstrained residual recovery without sacrificing recovery. Thus, the diagnostic result is not merely that refusal can be bypassed, but that non-refusal behavior remains recoverable while the active clamp is enforced and the monitored feature state stays close to its post-clamp value.

This pattern is not specific to the main AdvBench slice or to an obviously underspecified feature set. Appendix D repeats the protocol on HarmBench-Test and obtains 43/43 non-refusal recovery under low defended-feature drift. Appendix I sweeps broader refusal-feature sets and recomputes the strict-valid set for each clamp; recovery remains high in the stable operating range ($K = 5\text{--}20$), reaching 42/45 even at $K = 20$.

Attributing the recovery path. We next localize where the recovered behavior is carried. For each optimized perturbation, we replay four components under the same active clamp: clamped-feature changes, non-clamped SAE-feature changes, top- k non-clamped feature changes, and the SAE reconstruction residual. This tests whether recovery reopens the clamped refusal features, compensates through other visible SAE latents, or instead uses the SAE-unexplained residual channel. Appendix J gives the decomposition details.

Figure 6 identifies the SAE reconstruction residual as the dominant carrier. Residual replay nearly matches full recovery, while clamped-feature replay fails and non-clamped SAE-feature replay remains weak. Removing the top- k non-clamped feature component also preserves most recoveries, ruling out compensation through a small set of alternative SAE latents. Therefore, the refusal clamp blocks an interpretable feature-level route, but behaviorally sufficient information remains in the SAE-unexplained residual channel.

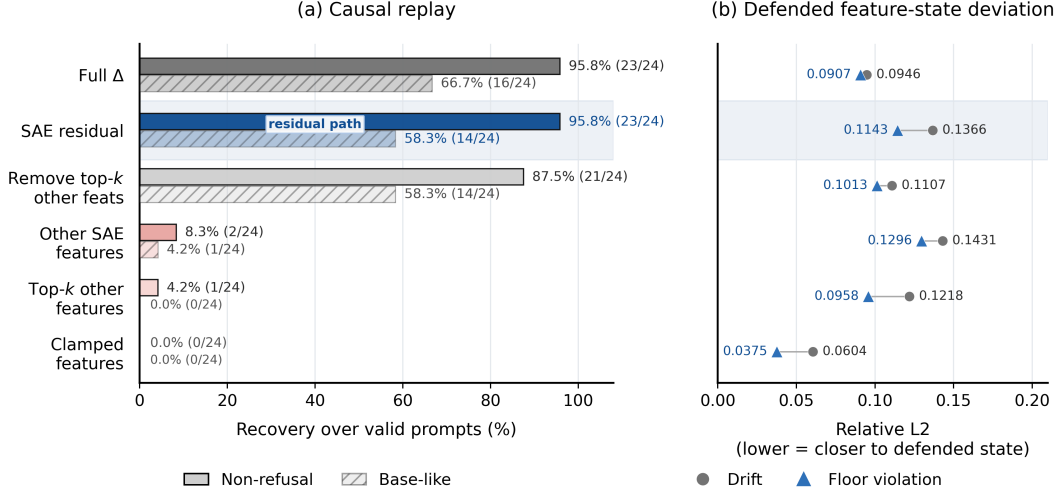


Figure 6: **Recovery is carried by the SAE reconstruction residual.** Replaying only the SAE residual nearly matches full recovery, while clamped-feature and non-clamped-feature replays largely fail, indicating that recovery is not explained by reopening visible SAE latents.

7 Discussion

Causal handles are not complete bottlenecks. Our experiments separate two notions that are often conflated in SAE-based intervention work. A selected feature set can be a useful causal handle: clamping it changes model behavior and can induce suppression, refusal, or loss of task performance. However, this does not imply that the same feature set is a complete bottleneck for the behavior. Across the settings we study, behavior can re-emerge from the defended residual state even when the original intervention remains active and the defended features remain close to their clamped values. Thus, feature-level intervention success should not by itself be interpreted as behavioral elimination.

The SAE residual is not inert error. The refusal attribution results suggest that the SAE reconstruction residual can carry behaviorally sufficient information that is invisible to the selected SAE feature basis. This changes how reconstruction error should be interpreted in safety-critical intervention settings. Even if the residual is small or treated as an error term for reconstruction, it may still contain computationally useful degrees of freedom through which the model can route around a feature clamp. The failure mode is therefore not merely reactivation of the clamped features or compensation by nearby SAE latents, but recovery through representational components not controlled by the intervention.

Implications for SAE-based safety evaluation. Our results do not imply that SAEs are useless for safety. Sparse features remain valuable for diagnosis, mechanistic localization, and local causal editing. The limitation is over-reliance: a defense that treats a selected SAE feature set as an exclusive behavioral bottleneck can be brittle. We therefore suggest that SAE-based defenses should be evaluated not only by whether a clamp suppresses a behavior, but also by whether the defended state is robust to constrained post-intervention recovery.

What stronger defenses would need. Simply enlarging the clamped feature set may improve coverage, but it does not directly address recovery through SAE-unexplained residual directions and may introduce capability or over-refusal side effects. Stronger defenses would need to constrain a broader portion of the computation: for example, by monitoring residual channels, using multi-layer or trajectory-level constraints, or explicitly training interventions against post-clamp recovery objectives. More generally, robust latent-space defenses should distinguish between blocking one interpretable route to a behavior and eliminating all recoverable routes to that behavior.